MIL-B-36446A
24 September 1973
SUPERSEDING
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22 October 1965

MILITARY SPECIFICATION

BURS, DENTAL SURGERY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dental surgery burs suitable for use in dental surgical procedures.

1.2 Classification. Burs shall be of the following classes, as specified (see 6.1):

Class AL - Angle handpiece, latch
Class S - Straight handpiece

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

Federal

FED-STD-151 Metal, Test Methods

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 OTHER PUBLICATIONS. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18-67 Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

E 92-72 Standard Method of Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Material. The burs shall be made of an alloy steel containing not less than 1.0 percent carbon and 1.0 percent tungsten.

3.2 Figures. The figures forming a part of this specification are specification control figures. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper design, and function, fit, assembly or operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

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3.3 Style, design and dimensions. The burs shall be of the Henahan design, size number 41. Class AL burs shall be suitable for use in latch-type, contra-angle handpieces. Class S burs shall be suitable for use with straight handpieces. Burs shall be suitable for cutting bone tissue in dental surgical procedures. Burs shall be of one-piece construction, consisting of a shank, a neck and a head. The burs shall conform to Figure 1 and 2 for style, design and dimensions, for the applicable class specified. The overall length of class AL burs shall be 1-1/8 $\pm$ 1/32 inches, and the overall length of class S burs shall be 2-1/2 $\pm$ 1/32 inches.

3.4 Hardness (Vickers test). When tested in accordance with 4.4.2, the hardness of the bur heads shall be no less than number 750 on the Vickers scale (Diamond Pyramid hardness). The bur necks shall be properly tempered to resist fracture or bending when in use.

3.5 Runout. The total indicated runout of the neck of the burs shall not exceed 0.003 inch, when measured as specified in 4.4.3.

3.6 Toughness. Burs shall show no sign of fracture, when tested as specified in 4.4.4.

3.7 Finish. The bur heads shall be sharp, cleanly cut and trimmed. The shanks and necks shall be smoothly finished and shall have a bright polish. Burs shall be free of defects such as cracks, nicks, slivers, pits, chips and other defects.

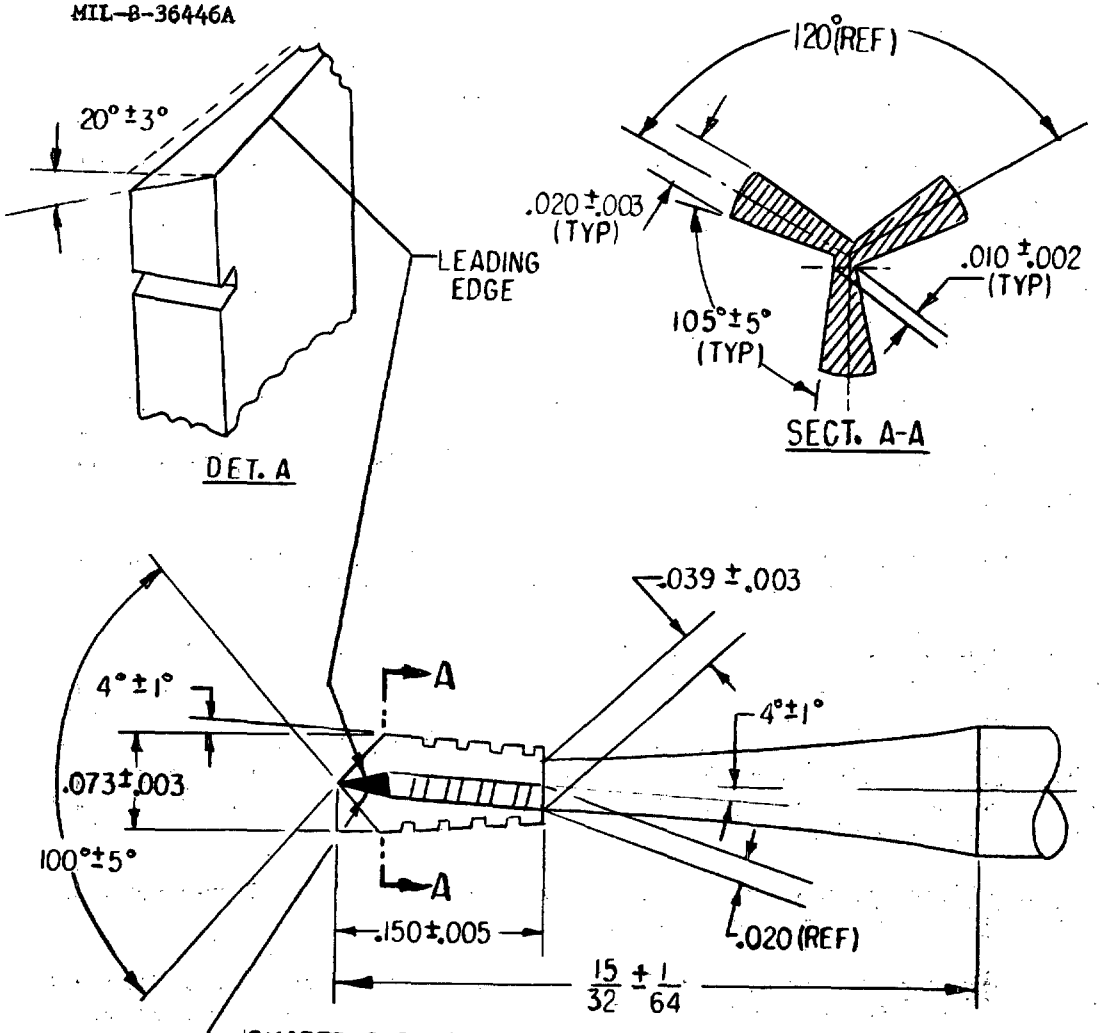
3.8 Identification marking. Marking of the bur shank with the name or registered trademark of the manufacturer is optional.

3.9 Workmanship. Workmanship shall be first class throughout. The burs shall be free from defects which may detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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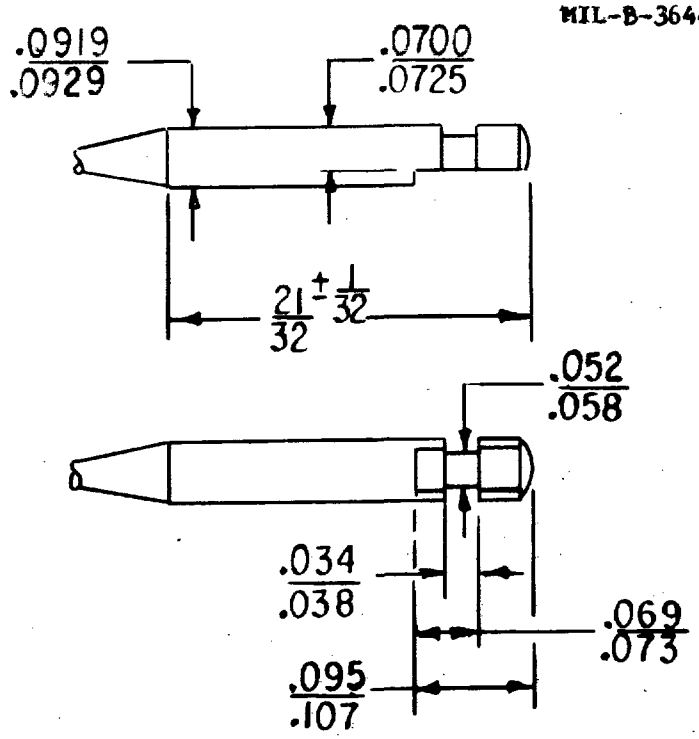
NOTES:

1. TYPICAL FOR CLASS-AL AND CLASS-S. SEE FIG. 2 FOR SHANK ENDS.
2. ALL DIMENSIONS ARE IN INCHES.

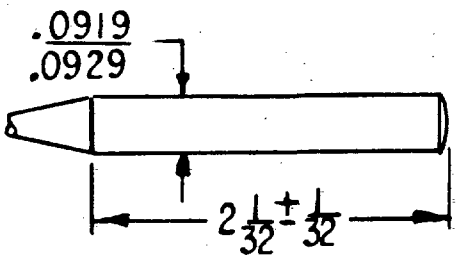
FIGURE 1

BUR, DENTAL SURGERY, HENAHAN, SIZE NO. 41.

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CLASS AL



CLASS S

NOTE:

ALL DIMENSIONS ARE IN INCHES

FIGURE 2

DIMENSIONS OF SHANK END OF BUR

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in Tables I and III (see 4.2.1 and 4.3) and Table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in Table I. The unit of product for examination purposes shall be one bur.

Table I. Examination Criteria

Type Defect	Inspection Level	AQL (Percent Defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

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4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table II, except that the acceptance number shall be zero for all sample sizes. Unit of product shall be one bur.

Table II. Test Criteria

Characteristic	Requirement	Test Procedure	Inspection Level
Chemical Analysis	3.1	4.4.1	(See 4.2.3)
Hardness	3.4	4.4.2	S-2
Runout	3.5	4.4.3	S-2
Toughness	3.6	4.4.4	S-2

4.2.3 Sampling of raw materials. In the absence of certificates of quality for the raw material, a sufficient quantity of each lot of raw material used in the fabrication of the burs shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The burs shall be examined to determine compliance with the requirements contained in this specification. Examination shall be conducted as specified in 4.2.1. Any bur in the sample containing one or more defects shall be rejected, and if the number of defective burs in any sample shall exceed the acceptance number for that sample, the lot represented by the sample shall be rejected. The burs shall be examined for defects including, but not limited to, those listed in Table III.

Table III. Classification of defects

Categories	Defects
Major A:	
101	Class not as specified.
102	Style and design not as specified.
103	Shank diameter not as specified (classes AL and S).
104	Latch end dimensions not as specified (class AL).
105	Burs not properly cut; not sharp and free from cracks, nicks, slivers, pits or chips.
106	Finish of bur heads not as specified.

Cont'd

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Major B:	
151	Dimensions of head not as specified.
152	Latch end (class AL) not cleanly machined.
Minor A:	
201	Necks and shanks not smoothly finished and brightly polished.
202	Packaging, packing and marking not as specified.
Minor B:	
251	Dimensions (other than 103, 104, and 151) not as specified.

4.3.1 Dimensions. Examination of dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with them.

4.3.2 Microscopic examination. Examination of bur defects not readily perceptible to the naked eye shall be made using a microscope at 10 power (10X) magnification.

4.4 Test procedures.

4.4.1 Chemical composition. The samples, selected as specified in 4.2.3, shall be analyzed in accordance with Method 111 or 112 of FED-STD-151. In case of dispute, Method 111 shall be used for referee analysis.

4.4.2 Vickers test. The Vickers hardness test shall be made in accordance with American Society for Testing and Materials (ASTM) Standard E92-72.

4.4.3 Runout. The bur shall be placed and held in a "V" block or precision chuck; and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow-graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360-degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

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4.4.4 Toughness. Blades shall show no sign of fracture when drilling carbon steel on a dental engine operating at 8,000 revolutions per minute (r.p.m.) under more than normal pressure. The carbon steel specimen shall have a minimum thickness of 0.30 inch, and a Rockwell hardness of B80 ± 5 when tested in accordance with ASTM Method E18-67. The duration of each test shall be one minute.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package, containing six (6) burs, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	1152 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit (6 burs) shall be inserted into a plastic holder in such a manner that burs will not contact one another. The burs shall be further protected with either a clear plastic cover which shall be secured with tape, or a paperboard cover. Containers shall be further packaged into a moisture-resistant cellophane bag and heat sealed.

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5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, class domestic. Commercial colors are acceptable on intermediate cartons and closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with paragraph 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

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5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with an asterisk to indicate where changes from the previous issue have been made.

6.3 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

Stock No.	Item Identification
6520-00-509-5050	BUR, DENTAL SURGERY, Angle Handpiece, Steel, Henahan, No. 41, 6s.
6520-00-509-6050	BUR, DENTAL SURGERY, Straight Handpiece, Steel, Henahan, No. 41, 6s.

6.4 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

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Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
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Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1363

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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